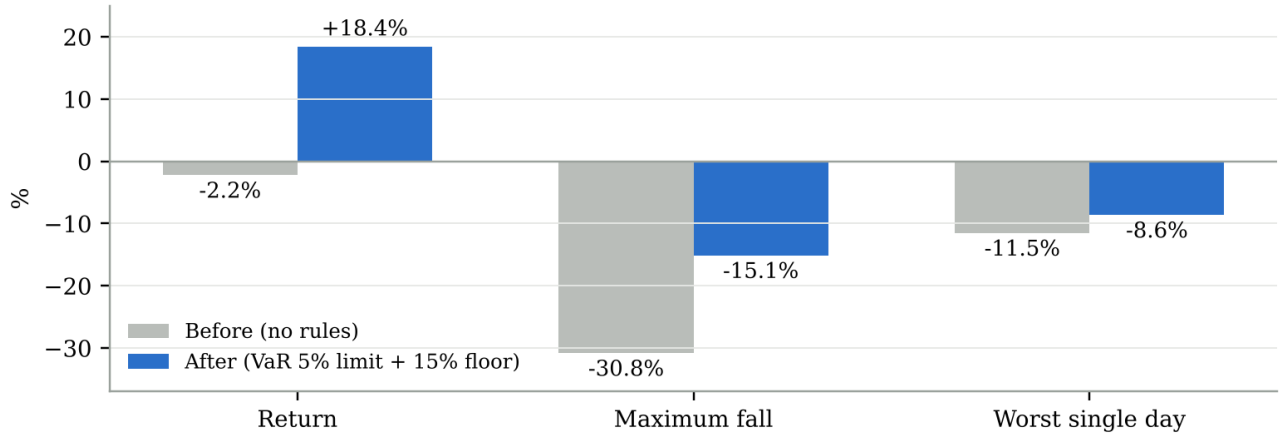
Portfolio Risk Case Study: Protecting Gains in a Leveraged Equity Account

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A private client's leveraged long/short US equity account was up about **60%** by mid-June, then gave most of it back by October; its 15% drawdown rule was never enforced. I rebuilt daily positions from the trade journal and measured risk each evening with **historical VaR(95)** computed from the actual holdings (500- and 250-day windows, the worse of the two). That traced the two largest losses to **leverage** (1.52× equity across 17 correlated AI and chip stocks; next day −10%) and **concentration** (one stock at 53% of equity; next day −11.5%), with VaR flagging the build-up seven weeks early. A **10,000-path Monte Carlo** stress test and a replay of 15 rule combinations led to two recommendations: **size the book each evening to a 5% VaR limit** (limit ÷ current VaR, capped at 100%) and **stop trading at a hard 15% loss floor**.

Replayed on the account's own record (2 Feb – 5 Oct), these rules turned **−2.2% into +18.4%**, cut the maximum drawdown from **−30.8% to −15.1%** and the worst single day from −11.5% to −8.6%. The Monte Carlo showed the early-June book had a **50%** chance of hitting the floor within 20 days, cut to **28%** by trimming size, and the floor lowered the chance of a fall over 20% from **31% to 3%**.



Before: the account as it was actually traded. After: the same trades replayed with the two rules, using only information available on each day.

What I did

1. **Cleaned the record.** Reconciled an incomplete trade journal (missing exits, typos, stock splits) and left out 7 disputed trades.
2. **Rebuilt daily positions** and exposure for every trading day from the clean trades.
3. **Measured risk each evening** with historical Value at Risk (VaR): "on a bad day, the 1-in-20 kind, how much could today's holdings lose?" It uses each holding's real past returns over the last one and two years.
4. **Found the causes** of the two largest losses.
5. **Tested 15 rule combinations** on the account's history, plus 10,000 simulated future paths (Monte Carlo) to estimate the odds of a large fall.

What went wrong

When	What the account looked like	Risk (VaR)	Next day
Early June	Positions worth 1.5× the account, spread over 17 closely linked AI and chip stocks: **too much leverage**	5.9%	−10%

Late June	One stock made up 53% of the account and half of all its risk: **too concentrated**	6.7%	−11.5%

The warning came early: the risk measure first rose above 4% in mid-April, **seven weeks before the first big loss**, while positions ran at 1.2–1.8× the account using margin.

July to October was a different problem: **overtrading**. Same-day round trips rose to 53% of trades and trading volume rose 29%. VaR only sees positions held overnight, so it cannot catch this, but a loss floor can.

The two rules

1. Size limit. Each evening, scale positions down so that a 1-in-20 bad day would cost no more than 5% of the account. In early June this would have cut the positions to about 85% of their size.

2. Loss floor. Stop trading when the account falls 15% below its highest value. This locks in gains made earlier.

Before versus after

Rule set (2 Feb – 5 Oct)	Return	Maximum fall	Worst day	Sharpe
Before: no rules	−2.2%	−30.8%	−11.5%	0.06
Loss floor only	+19.0%	−15.8%	−11.5%	0.94
Size limit 4% + floor	+13.7%	−15.5%	−6.9%	0.86
After: size limit 5% + floor	**+18.4%**	**−15.1%**	**−8.6%**	**1.01**

2 Feb is the first day with a complete position record. Measured over the full year from 1 January, the chosen rules lift the return from +13.0% to +36.6% and the Sharpe ratio from 0.62 to 1.57.

What the simulations showed

- With the early-June positions, the account had a **50% chance** of falling 15% from its peak within the next month. Trimming positions to 68% of their size (a 4% limit) cut that to **28%**.
- Adding the loss floor cut the chance of a fall of more than 20% from **31% to 3%**.

Takeaways

- **The loss floor protects the year**; the size limit softens the worst days. Each covers a gap the other leaves.
- **Tighter is not better.** Limits of 2–3% cut the momentum runs this trader earns from.
- **Good results came from fewer, smaller bad days**, not from avoiding the rally: the rules kept almost all of the April–June gain.

Notes

These are replayed (backtested) results on one account over one year, with one major drawdown, so the exact percentages are indicative. All 15 rule combinations were reviewed; the chosen rules are not the best on every measure. Client details are withheld.